Project # 5

Mobile Repair Shop Tracker

Software Requirement Specification

Website Name: JMW Fix Track

Submitted by:-

Student Name: Javeria Sarfraz

Discipline: BS IT

Introduction:-

The mobile repair shop tracker is a software system designed to manage the daily activities of a mobile repair shop. It helps the shop owner record customer information, register repair jobs, assign technicians, and manage payments.

The system provides an organized way to track mobile phones during the repair process. Customer can also check the status of their repairs using a repair job number. This system reduces paper work, saves time and improves the efficiency of the repair shop.

Problem Statement:-

Many mobile repair shops manage their records manually. This can cause problems such as lost customer records, difficulty tracking repair jobs, delays in updating repair statuses, and errors in payment records.

The JMW Fix Track is proposed to solve these problems by maintaining customer, technicians, repair, and payment information in an organized system.

Requirement Analysis:-

The requirement analysis is the process of gathering, understanding, and describing the behavior, the features, the needs of users and performance, expectations of the JMW Fix Track web application. The goal is to understand what the system does, what features are essential for users and how it should work. It includes both functional and non-functional requirements of the system. These requirements help to ensure the final product meets user expectations and supports JMW Fix Track management activities efficiently.

User Classes and Characteristics:-

User Classes	User Characteristics
Customers	Customers are those who will use JMW Fix Track to register online, view mobile repair services and prices, book repair services online, and track their repair status. They prefer a simple, and user friendly interface that allows them to manage workouts, track their status and make payments easily from any device.
Technicians	A skilled mobile repair technician is responsible for diagnosing and repairing mobile phones, communicating directly with customers, and updating repair progress and status on the website.
Admin	The admin manages the overall platform. Admin oversees repair bookings, assign repair jobs to technicians, manages services and prices, and maintain customer records and billing information. They ensure that repair statuses are updated and the website operates smoothly.

Requirement Identifying Techniques:-

For the JMW Fix Track web application, the Use Case technique is selected to identify and organize system requirements. Use cases help to define the behavior and interactions between users and the system. Each use case describes a specific action that a user can perform, making it easier to understand the system's functional needs. Detailed use case description and a use case diagram will be used to show the flow of user actions and system responses.

Functional Requirements:-

Functional requirements are the requirements that tell us what the system does. It defines the features, functions, and services of the website that user needs. Some functional requirements of the website JMW Fix Track are:-

Sign Up (FR-1)

Identifier	FR-1
Title	Sign Up
Requirement	The system should provide the access to create an account to all users to access their respective features.
Source	Customers, Technicians, and Admins
Rationale	The system will allow all users to create their accounts to access the website.
Business Rule	Email and password must be unique
Pre-Condition	Valid email, username and password
Post-Condition	Sign in to their existing account

Sign in (FR-2)

Identifier	FR-2
Title	Sign in

Requirement	The system should provide the facility to access to the existing account of the user.
Source	Customers, Technicians, and Admins
Rationale	The proposed system will allow all users to login to their existing accounts.
Business Rule	Email and password must be unique
Pre-Condition	Sign up (User should have an account)
Post-Condition	Registration to submit repair order

New Repair Job Form (FR-3)

Identifier	FR-3
Title	New repair Job Form
Requirement	The website will provide a registration form to the customers to add their device details in it after which system will provide a unique job number to the customer.
Source	Customers
Rationale	The system will provide a form to the customers to submit their repair order.
Business Rule	The customers will provide their information and fault details of their devices.
Pre-Condition	Sign in
Post-Condition	Unique job number assignment

Unique Repair Job number Assignment (FR-4)

Identifier	FR-4
Title	Unique Repair Job Number Assignment
Requirement	When a customer will submit a repair order, a unique job number will assign to him and the record of that customer along his unique job number will beaded to the database.
Source	Customers
Rationale	Customers can easily be identified using their job number.
Business Rule	Job number assignment must be handled by the admin. He can edit and view job numbers.
Pre-Condition	Submit the repair order by filling the repair job form.
Post-Condition	Assign jobs to technicians

Assign Jobs to Technicians (FR-5)

Identifier	FR-5
Title	Assign Jobs to Technicians
Requirement	The system will allow the admin to assign repair jobs to available technicians through the admin panel.
Source	Admin
Rationale	To assign a repair job to technician and ensure that the technician will do their job responsibly and complete on time.
Business Rule	Technician should repair the order on time.
Pre-Condition	Unique job number assignment
Post-Condition	Update Repair Status & Status Tracking

Update Repair Status & Status Tracking (FR-6)

Identifier	FR-6
Title	Update Repair Status & Status Tracking
Requirement	The system will allow technician to update customer's repair status and customers will track their mobile repair status. Admin can also handle the repair status.
Source	Admin, Customers and technicians
Rationale	To keep customers informed about repair progress.
Business Rule	An Alert will send to the customers when their tasks are completed.
Pre-Condition	Assign jobs to technicians
Post-Condition	Generate Bill & Receive Payment

Generate Bill & Receive Payment (FR-7)

Identifier	FR-7
Title	Generate Bill & Receive Payment
Requirement	The admin will hand over the mobile to the customer, generate the bill and receive the payment.
Source	Admin
Rationale	Trusted payment methods for customers
Business Rule	Payment will be recorded in database and bill will only generate for completed order.
Pre-Condition	Update Repair Status & Status Tracking
Post-Condition	Recorded in database

Admin Panel (FR-8)

Identifier	FR-8
Title	Admin Panel
Requirement	The system will allow admin to view customers records, manage their job numbers, track their repair status, assign jobs to technicians, generate bills and receive payments.
Source	Admin
Rationale	Allows management of entire shop in one place.
Business Rule	Only the admin should be able to access and modify any data.
Pre-Condition	Login to his account
Post-Condition	Manage Database records and whole website

Jobs List by Status (FR-9)

Identifier	FR-9
Title	Jobs List by Status
Requirement	Admin can view all jobs lists by their repair status.
Source	Admin

Rationale	Easy to manage the jobs
Business Rule	The jobs having same repair status will show in the same ist.
Pre-Condition	Access to the admin Panel
Post-Condition	Track repair statuses.

Track Repair by Job Number (FR-10)

Identifier	FR-10
Title	Track Repair by Job Number
Requirement	The customers can track their repair status by their unique assigned job number.
Source	Customers
Rationale	Easy to Track their status
Business Rule	Customers can see only their own status.
Pre-Condition	Login to their account
Post-Condition	Communicate with admin or technician if they face any issue.

Revenue Report (FR-11)

Identifier	FR-10
Title	Revenue Report
Requirement	The admin can track his shop's daily, weekly or monthly income report.
Source	Admin
Rationale	To track income details and maintain payment records.
Business Rule	This report will be generated based on received payment.
Pre-Condition	Receive Payments
Post-Condition	Record and payment management

Non-Functional Requirements

1. Reliability

The website should provide reliable and accurate results. Data should not be lost. It should not contain errors and failures while using it.

2. Usability

The website should contain user friendly interface. It will be easy to use for customers and technicians. It should provide clear and user friendly features. All functions and operations should be easy to use and understandable to the users.

3. Performance

The performance of the website should be efficient and fast. Webpages should be easily loaded and requests should be responded quickly. Repair booking, job number assignment, repair status updates and billing operations should be perform smoothly.

4. Security

The website must contain security for all user accounts. It should provide a secure login and registration to all users and customers. Customers can only see their status and data on the website. It will be secured from unauthorized access.

5. Compatibility

The website will be compatible for all types of devices such as mobile phones, tablets, laptops etc. It can easily be loaded on any type of browser such as Google Chrome, Mozilla and Firefox etc. Users can easily access all features on any device.

6. Maintainability

The website will be well-organized and easy to maintain. It can be modify, update, and update in future when needed. Errors can easily be identified and removed to improve the website and make it better.

7. Data Backup

The website should take the regular backup of data in database to prevent data loss. It can be helpful in case of accidentally loss data. Backup means making the copy of data.

8. Scalability

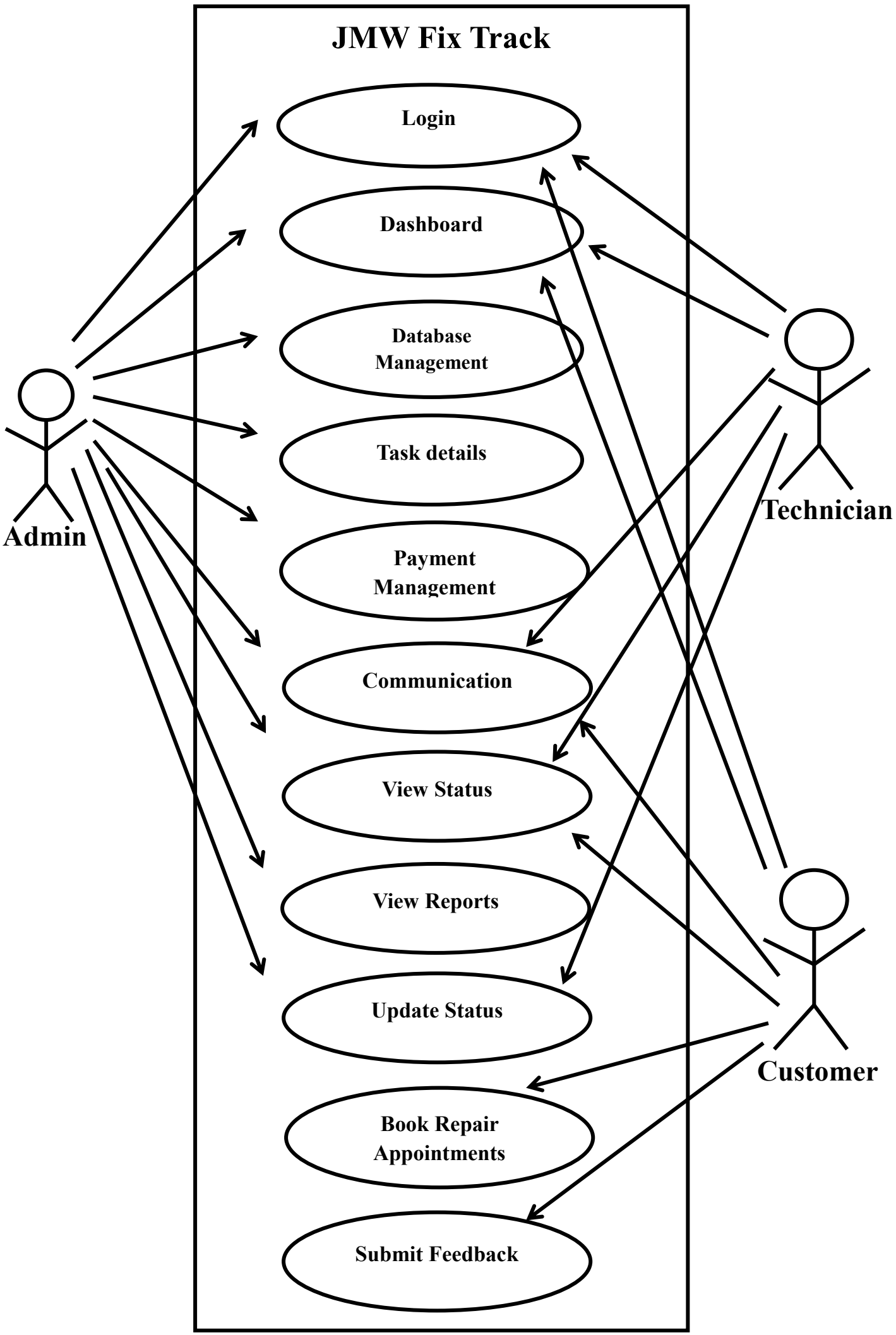
After some time when the number of users will increase, the website will be easy to use and compatible for all users. It should perform well for all users.

Use Case Diagrams:-

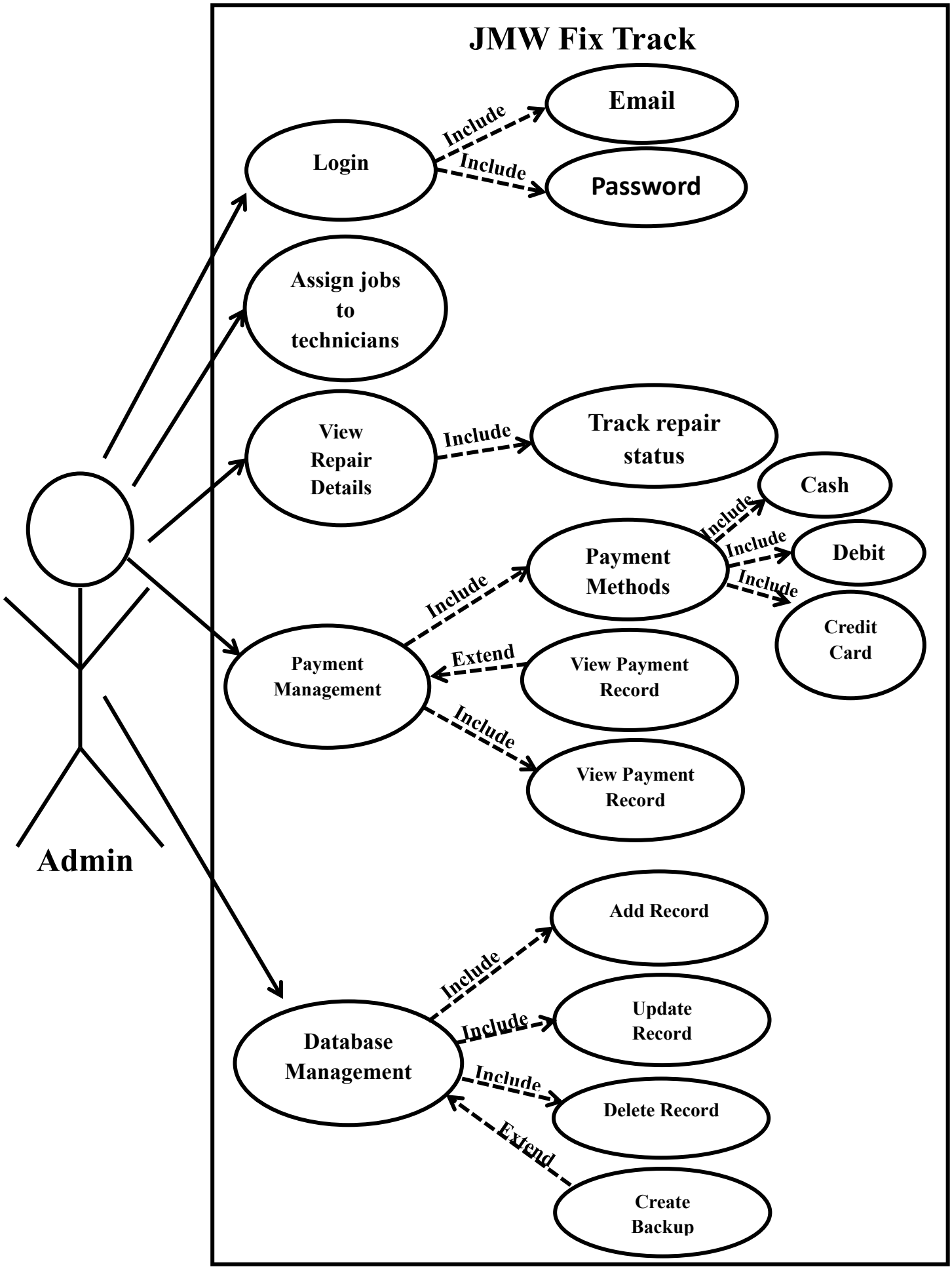
A use case diagram is a diagram that represents the relationship between users and the system. It defines the activities that different users can perform to interact with the system. It defines which functions can be performed by the users.

Use Cases of JMW Fix Track:-

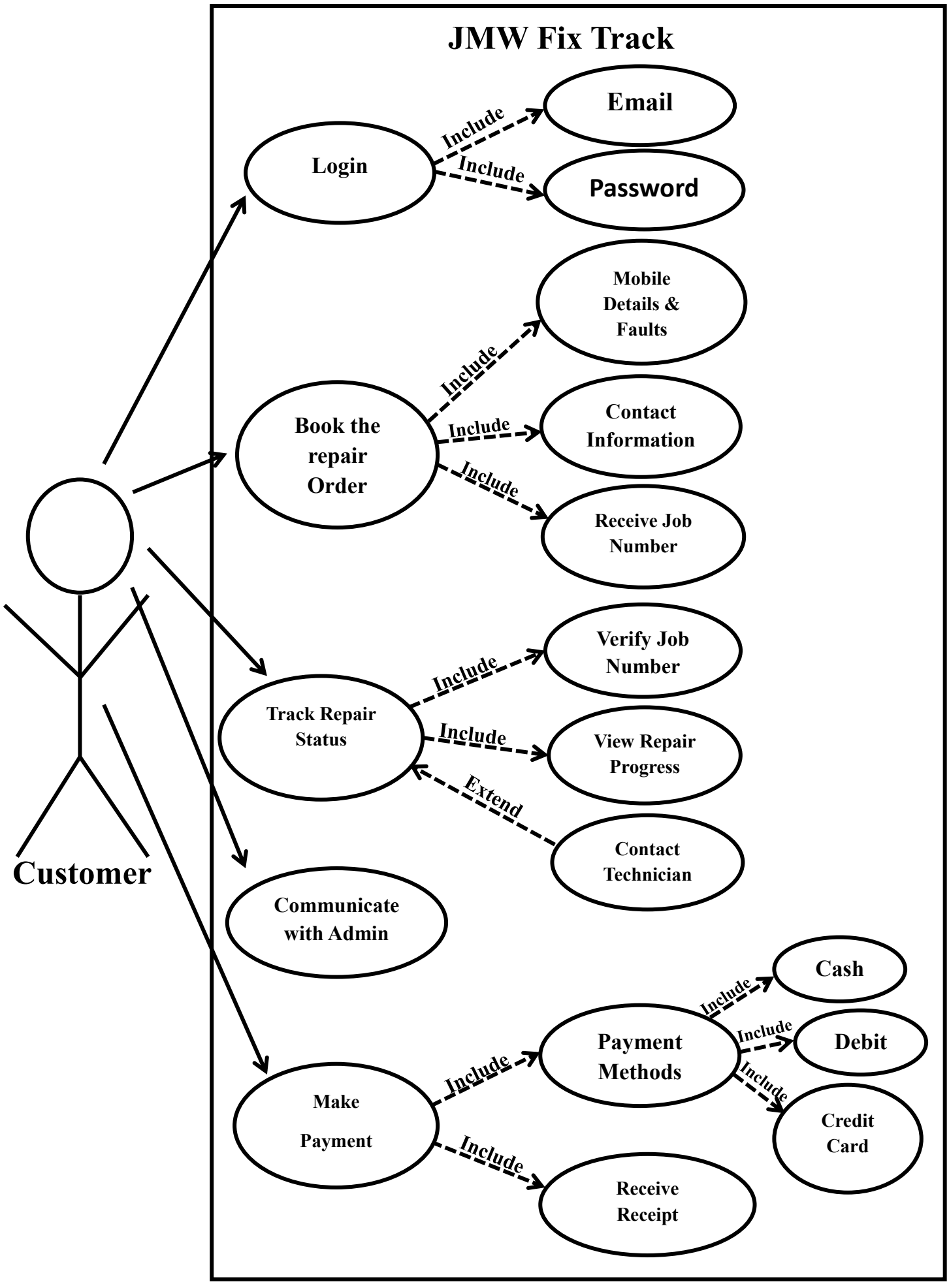
This website will interact with three types of users (Customers, Admin and Technicians). Use case diagram of these users are given below:-


JMW Fix Track
Login
Dashboard
Database Management
Task details
Payment Management
Communication
View Status
View Reports
Update Status
Book Repair Appointments
Submit Feedback
Admin
Technician
Customer

1. **Admin**


JMW Fix Track
Email
Login
Include
Include
Password
Assign jobs to technicians
View Repair Details
Include
Track repair status
Cash
Include
Payment Methods
Include
Debit
Include
Credit Card
Include
Payment Management
Extend
View Payment Record
Include
View Payment Record
Admin
Add Record
Include
Database Management
Include
Update Record
Include
Delete Record
Extend
Create Backup

2. Customers


JMW Fix Track
Email
Login
Include
Include
Password
Mobile Details & Faults
Include
Book the repair Order
Include
Contact Information
Include
Receive Job Number
Verify Job Number
Include
Track Repair Status
Include
View Repair Progress
Extend
Contact Technician
Customer
Communicate with Admin
Cash
Include
Include
Payment Methods
Debit
Include
Credit Card
Include
Make Payment
Include
Receive Receipt

3. Technicians

